MILITARY SPECIFICATION

GRENADE, GP: M36,

PARTS FOR

1. SCOPE

1.1 This specification covers the parts for one type of grenade designated as Grenade, GP: M36.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

PURCHASE SPECIFICATION

AEI-170	-	Primer, Percussion, M98, Parts for and Loading, Assembling and Packing

FSC: 1330

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

F-8865574	-	Box Fiber, Inner, for Housing Assy, XM36E2 or XM36E4
C-8865575	-	Box Fiber, Outer, for Housing Assy, XM36E2, XM36E4
C-9211946	-	Grenade, GP: M36
D-9211947	-	Housing Assembly
D-9211966	-	Grenade Body Assembly
C-9211968	-	Adapter, Tube
D-9211971	-	Cap, Retainer

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PUBLICATIONS

ARMY

ET-9211947 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision).")

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117-64 - Standard Method of Salt Spray (Fog) Testing (Book 31)

ASTM E 18-67 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials (Book 31)

ASTM B 8-69 - Tension Testing of Metallic Materials (Book 31)

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

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3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The grenade parts shall comply with all requirements specified on Drawing (Dwg) D-9211947, C-9211968, D-9211971, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Simulated grenade catapult test.-The simulated grenade, consisting of a ball to simulate the actual weight and size as specified on Dwg D-9211966, and propellant charge and housing assemblies as specified on Dwg. D-9211947, shall exhibit an average impact velocity of 125 plus or minus (±) 10 feet per second (fps) without evidence of cracking or complete splitting of housing when tested as specified in 4.4.17.

3.4 Mechanical properties of housing.-Cold-working or heat-treating (including quenching-and-tempering, normalizing, normalizing-and-tempering, annealing, and stress-relieving) operations, shall be performed in such a manner as to produce the specified properties uniformly throughout the entire lot of the housings. The contractor shall determine the detailed procedures to produce parts meeting the required mechanical properties. These procedures shall be documented and submitted with the first article sample to the Contracting Officer. (See 4.2.5) Changes in these procedures shall also be documented and submitted to the Contracting Officer.

3.5 Workmanship.-All parts and assemblies shall be fabricated, finished and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.6)

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

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4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2 (See 6.1). The first article sample shall consist of:

a. Seventy-five (75) of each part, sub-assembly and assembly.
b. Seventy-five (75) housing assemblies (Dwg. D-9211947).
c. Ten (10) housings, after heat treatment, (prior to machining for forgings; after machining of bore and prior to machining slider cavity for bar stock extrusions).

All parts and assemblies shall have been produced by the contractor or furnished by a supplier using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance for materials. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract.and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contractig Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification, and any or all requirements of the applicable drawings. In addition, any or all of the following examinations and tests will be performed as specified in 4.4 using equipment in accordance with 4.3.4.

WARNING: The slider assembly contains an explosive component Safety precautions shall be observed.

4.2.2.1 Hardness of slider stop pin, yoke pin and hinge pin (See Dwgs B-9211965, B-9211954 and C-9211950, resp.-Seventy-five (75) each part shall be subjected to this test as specified in 4.4.1.

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4.2.2.2 Tensile strength of housing (prior to machining slider hole) (See Dwg. F-9211948).-One (1) specimen shall be taken from each of the ten (10) housings and prepared and tested as specified in 4.4.2.

4.2.2.3 Permanent set of housing (prior to machining slider hole) (See Dwg. F-9211948).-The ten (10) housings shall be subjected to this test as specified in 4.4.3.

4.2.2.4 Push-out test of retainer cap (See Dwg. D-9211971).-The seventy-five (75) retainer caps shall be subjected to this test as specified in 4.4.4.

4.2.2.5 Load tests of slider spring (after temperature-conditioning) (See Dwg. C-9211962).-The seventy-five (75) slider springs shall be subjected to these tests as specified in 4.4.5.

4.2.2.6 Load test of springs of yoke (See Dwg. D-9211952).-The seventy-five (75) yokes shall be subjected to this test as specified in 4.4.6.

4.2.2.7 Load test of leaf springs of outer spring (See Dwg. F-9211944).-The seventy-five (75) outer springs shall be subjected to this test as specified in 4.4.7.

4.2.2.8 Push test of side tabs of outer spring (See Dwg. F-9211944).-The seventy-five (75) outer springs shall be subjected to this test as specified in 4.4.8.

4.2.2.9 Direction of grain of yoke and outer spring (See Dwgs. D-9211952 and F-9211944).-Upon completion of tests specified in 4.2.2.6, 4.2.2.7 and 4.2.2.8, five (5) yokes and outer springs shall be subjected to this test as specified in 4.4.9.

4.2.2.10 Leakage test of diaphragm (See Dwg. C-9211963).-The seventy-five (75) diaphragms shall be subjected to this test as specified in 4.4.10.

4.2.2.11 Air flow test of restrictor (See Dwg. C-9211964).-The seventy-five (75) restrictors shall be subjected to this test as specified in 4.4.11.

4.2.2.12 Push test of firing pin of yoke assembly (See Dwg. C-9211951).-The seventy-five (75) yoke assemblies shall be subjected to this test as specified in 4.4.12.

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4.2.2.13 Push test of primer in slider assembly (See Dwg. B-9211955).-The seventy-five (75) slider assemblies shall be subjected to this test as specified in 4.4.13. (WARNING this is a loaded assembly. All safety precautions shall be observed.)

4.2.2.14 Security of hinge pins and outer springs (See Dwg D-9211947).-The hinge pins and outer springs of the seventy-five (75) housing assemblies shall be subjected to this test as specified in 4.4.14.

4.2.2.15 Pull test of safety wire (See Dwg. D-9211947).-The safety wire of the seventy-five (75) housing assemblies shall be subjected to this test as specified in 4.4.15.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4 Additional first article inspection.

4.2.4.1 Slider arming time of housing assembly (See Dwg. D-9211947).-Seventy-five (75) housing assemblies shall be subjected to this test as specified in 4.4.16.

4.2.4.1.1 Rejection.-If two (2) or more housing assemblies fail to arm within the slider delay time specified on the applicable drawing, the first article quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new first article quantity, or portion thereof, as specified by the Contracting Officer. The contractor shall continue to submit new or supplemental quantities, as necessary, until such time as a quantity successfully passes the inspection specified, or until otherwise directed by the procuring activity.

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4.2.4.2 Simulated grenade catapult test (See 3.3).-Seventy-five (75) housing assemblies (See 6.5) shall be assembled with a Government-furnished container assembly, and a ball to simulate the weight and size as specified on Dwg. 8865585. Assembly shall be in accordance with Dwg. C-9211946. Instrumentation shall be provided to check impact velocity on every fifth sample shot. Velocity results shall be recorded (See 6.6). The test shall be performed asnspecified in 4.4.17.

4.2.4.2.1 Results to be discarded. Recorded data required by 4.2.4.2.2 shall be discarded if any one of the following occurs:

a. Impact velocity less than 100 feet per second (fps)
b. Equipment malfunction.
c. Poor orientation on impact.
d. External interference to ball while in flight.

(Additional grenades shall be provided to complete the sample quantity specified)

4.2.4.2.2 Records.-Record for information, the following:

a, Impact velocity.
b. Housing malfunctions.
c. Crushed or cracked housing.

4.2.4.2.3 Rejection.-If one or more grenades fail to comply with the specified requirements, the first article quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new first article quantity as specified by the contracting officer.

4.2.5 Mechanical properties.-Documentation detailing all equipment, operations, and procedures to be used during the term of this contract, and affecting one or more of the mechanical properties, shall be submitted as part of first article inspection. Changes in the established procedures or equipment during the course of the contract shall be documented and submitted, with test data, as specified in 6.6. Documentation shall include actual chemical and metallurgical analyses of material, including wrought materials, (hardness, grain size and grain structure with photographs). The finished items shall also provide data for tensile strength, yield strength, percent elongation, percent reduction in area and charpy v-notch impact tests. Documentation covering normalizing, tempering, stress-relieving and annealing, when any of these operations are used, shall provide the shortest and

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longest time at the temperature for each item, size and type of furnace or oven. When quenching is employed, documentation shall indicate the quenching medium, its temperature, method of presentation of items into the medium and length of time in the medium. This documentation, with actual test data, shall be signed by a responsible agent, of the prime contractor and submitted to the technical agency specified in 6.6.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Housing (See Note 1)
b. Outer spring
c. Primer (See 4.3.3.5)

NOTE 1: For housings, one lot interfix number from one manufacturer requires the lot interfix number to be related to the aluminum alloy material that is produced by one manufacturer in one unchanged process, in accordance with same drawing, same drawing revision, same specification and same specification revision.

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4.3.1.1 Housing heat treatment lot.-A heat treatment lot shall consist of not less than two hundred (200) housing of one lot of specified aluminum alloy that has been heat-treated by the same process, in the same furnace, at the same temperature, and for the same length of time. All lot identification control must be maintained until all mechanical properties tests have been completed. The method of control must be approved by the Government.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. In addition, at the option of the procuring activity, defect characteristics of the same classification which have been separated into two (2) groups to specifically allow the optional method of inspection in accordance with 4.3.3.1, 4.3.3.2, 4.3.3.3, and 4.3.3.4 may be combined into one (1) group when the option of 4.3.3.1, 4.3.3.2, 4.3.3.3 and 4.3.3.4 is not exercised. The AQL for the combined grouping will be as indicated in the applicable classification of defects paragraph. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Slider (See Dwg D-9211956 covering a detail of Dwg. 9211955.-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major	AQL 1.50 Percent (See 4.3.3.1)		
101.	Total length	Gage	01001
102.	Largest outside diameter	Gage	01002
103.	Diameter at rounded end	Gage	01003
104.	Length of diameter at rounded end, min	Gage	01004

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105.	Outside diameter at flat end	Gage	01005
106.	Depth to bottom of stop slot	Gage	01006
107.	Diameter of primer cavity flash hole in slot	Gage	01007
108.	Width of slot	Gage	01008
109.	Depth to bottom of primer cavity from crimping surface	Gage	01009
110.	Depth to bottom of primer cavity from outside diameter	Gage	01010
111.	Symmetry of slot with outside diameter	Gage	01011
112.	Perpendicularity of primer cavity with outside diameter including symmetry with slot	Gage	01012
113.	Length to end of stop slot from rounded end	Gage	01013
114.	Location of primer cavity	Gage	01014
115.	Location of safety hole	Gage	01015
116.	Distance to largest diameter shoulder from flat end	Gage	01016
117.	Concentricity of flash hole in slot with primer cavity	Gage	01017
Minor	*AQL 2.50 Percent (See 4.3.3.1)		
201.	Diameter of primer cavity, maximum (max)	Gage	01019
202.	Perpendicularity of safety hole with outside diameter including symmetry with stop slot	Gage	01020
203.	Concentricity of end diameters with largest outside diameter	Gage	01021
204.	Concentricity of second diameter (safety pin portion) with largest outside diameter	Gage	01022
205.	Diameter of safety hole, max	Gage	01023
206.	Outside diameter of safety hole portion	Gage	01024
Minor	*AQL 1.50 Percent		
207.	Radius missing or incorrect	Visual	01025
208.	Evidence of poor workmanship (See 3.5)	Visual	01026
209.	Finish improper	Visual	01027

* Combined AQL for Minor defects 201-209 inclusive is as follows:
AQL 4.00 Percent (See 4.3.2)

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4.3.2.2 Slider assembly (See Dwg. B-9211955).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Outside diameter, max (after primer load test)	Gage	02001
102.	Primer missing or inverted	Visual	02002
103.	Primer damaged	Visual	02003
104.	Slider damaged (cut, serrated, chip, nick, etc)	Visual	02004
105.	Burr, projection or raised metal	Visual	02005
106.	Stake missing or inadequate	Visual	02006
Minor	AQL 1.00 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	02008

4.3.2.3 Stop pin, slider (See Dwg. B-9211965).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Total length	Gage	03001
102.	Length of knurled section	Gage	03002
103.	Diameter over knurl, min	Gage	03003
104.	Diameter of plain end, min	Gage	03004
Minor:	None defined		

4.3.2.4 Spring, slider (See Dwg. C-9211962).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Solid height, max	Gage	04001
102.	Outside diameter, max	Gage	04002
103.	Inside diameter, min	Gage	04003

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Categories	Defects	Method of Inspection	Code No.
Minor	AQL 1.00 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	04004
202.	Ends not squared	Visual	04005

4.3.2.5 Cover, spring, slider (See Dwg. B-9211961).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent (See 4.3.3.3)		
101.	Outside diameter	Gage	05001
102.	Thickness through bottom, min	Gage	05002
103.	Inside diameter, min	Gage	05003
Minor	*AQL 1.00 Percent (See 4.3.3.3)		
201.	~~Diameter~~ of hole	Gage	05004
202.	Concentricity of hole with outside diameter	Gage	05005
Minor	AQL 0.65 Percent		
203.	Burr or projection on inside face	Visual	05006

*Combined AQL for Minor defects 201-203, inclusive, is as follows: 1.50 Percent (See 4.3.2)

4.3.2.6 Cap, retainer (See Dwg. D-9211971).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	**AQL 0.40 Percent (See 4.3.3.3)		
101.	Height of outer bend	Gage	06001
102.	Outside diameter	Gage	06002
Minor	*AQL 2.50 Percent (See 4.3.3.3)		
201.	Diameter of hole	Gage	06003
202.	Thickness	Gage	06004
203.	Depth to basic diameter	Gage	06005
204.	Concentricity of hole with outside diameter	Gage	06006

** See note next page.

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Minor	AQL 1.00 Percent		
205.	Evidence of poor workmanship (See 3.5)	Visual	06007
206.	Protective coating damaged (base metal exposed)	Visual	06008

NOTE: Characteristics 101 and 102 apply prior to heat treatment

*Combined AQL for Minor defects 201-206, inclusive, is as follows: AQL 2.50 Percent (See 4.3.2)

4.3.2.7 Pin, firing (See Dwg. B-9211953).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Profile of point	Gage	07001
102.	Concentricity of point with body	Gage	07002
103.	Diameter of body	Gage	07003
104.	Length from shoulder to tip of point, min	Gage	07004
105.	Point damaged (bent or flattened)	Visual	07005
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	07006

4.3.2.8 Wire, safety (See Dwg. B-9211960).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent (See 4.3.3.3)		
101.	Diameter of wire	Gage	08001
Minor	AQL 1.00 Percent (See 4.3.3.3)		
201.	Loop missing or incorrectly formed	Visual	08002
202.	Evidence of poor workmanship (See 3.5)	Visual	08003

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4.3.2.9 Yoke (See Dwg. D-9211952).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	*AQL 1.00 Percent (See 4.3.3.3)		
101.	Distance from inside bottom of tab to top of slotted tab	Gage	09001
102.	Distance from inside bottom of tab to top of slots	Gage	09002
103.	Width of bottom at inside datum	Gage	09003
104.	Symmetry of tab width with datum width	Gage	09004
105.	Symmetry of slots with tab width	Gage	09005
106.	Symmetry of firing pin hole with datum width and inside length	Gage	09006
107.	Diameter of firing pin hole, max	Gage	09007
108.	Width of tabs, min	Gage	09008
109.	Width of slots	Gage	09009
110.	Length of slots	Gage	09010
Major	AQL 0.40 Percent		
111.	Burr or projection on inside of part	Visual	09011
Minor	*AQL 1.00 Percent (See 4.3.3.3)		
201.	Thickness of tab	Gage	09012
202.	Width of angled sides	Gage	09013
Minor	AQL 1.00 Percent		
203.	Evidence of poor workmanship (See 3.5)	Visual	09014
204.	Protective coating damaged (bare spot or scratches)	Visual	09015

*Combined AQL for Major defects 101-111, inclusive, is as follows: AQL 1.00 Percent; combined AQL for Minor defects 201-204, inclusive, is as follows: 1.50 Percent (See 4.3.2)

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4.3.2.10 Yoke assembly (See Dwg. C-9211951).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major		AQL 0.65 Percent	
101.	Height of firing pin, max	Gage	10001
102.	Symmetry of firing pin diameter with reference width and width of tabs	Gage	10002
103.	Firing pin loose	Manual	10003
104.	Firing pin point damaged (bent or flattened)	Visual	10004
Minor		AQL 1.50 Percent	
201.	Height of staked head of firing pin, max	Gage	10005
202.	Firing pin head protrudes over radius of angled sides	Visual	10006
203.	Springs damaged (bent or flattened)	Visual	10007
204.	Evidence of poor workmanship (See 3.5)	Visual	10008

4.3.2.11 Spring, outer (See Dwg. F-9211944 covering a detail of Dwg. D-9211947).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Safety wire hook incorrectly formed (adjacent hooks fail to form a closed loop)	Visual	11001
Major		*AQL 1.00 Percent (See 4.3.3.3)	
101.	Depth to lip of largest safety wire hook, max	Gage	11002
102.	Depth to lip of larger of two adjacent safety wire hooks, max	Gage	11003
103.	Location of hinge pin hole	Gage	11004
104.	Location of safety wire slot	Gage	11005
105.	Flatness of face	Gage	11006
106.	Thickness of material	Gage	11007

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107.	Distance between hinge pin hole and locking tab slot	Gage	11008
108.	Diameter of hinge pin hole, max	Gage	11009
109.	Location of datum on dimple of hinge pin hole	Gage	11010
110.	Concentricity of datum on dimple with hinge pin hole	Gage	11011
111.	Width at datum height	Gage	11012
Major	AQL 0.65 Percent		
112.	Burr or projection	Visual	11013

*Combined AQL for Major defects 101-112, inclusive is as follows: AQL 1.50 Percent (See 4.3.2)

Minor	*AQL 2.50 Percent		
201.	Angle of sides	Gage	11014
202.	Distance from side to smallest safety wire hook	Gage	11015
203.	Distance from side to largest safety wire hook	Gage	11016
204.	Distance from hinge pin hole to edge of angled portion on face	Gage	11017
205.	Width of slot between forward and reverse safety wire hooks, max	Gage	11018
Minor	AQL 1.00 Percent		
206.	Protective finish inadequate	Visual	11019
207.	Evidence of poor workmanship (See 3.5)	Visual	11020

*Combined AQL for Minor defects 201-207, inclusive, is as follows: AQL 2.50 Percent (See 4.3.2)

4.3.2.12 Housing (See Dwg. F-9211948).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 2.50 Percent		
101.	Distance from top to top basic	Gage	12001
102.	Distance from top to largest outside basic	Gage	12002
103.	Distance from top to smallest outside basic near slider cavity	Gage	12003

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104.	Distance between yoke slots	Gage	12004
105.	Total height	Gage	12005
106.	Total width	Gage	12006
107.	Diameter of hinge pin holes below countersink, max	Gage	12007
108.	Diameter of yoke pin holes, max	Gage	12008
109.	Diameter of slider stop hole, max	Gage	12009
110.	Diameter of slider cavity	Gage	12010
111.	Diameter of large cavity	Gage	12011
112.	Diameter of flash hole	Gage	12012
113.	Distance between retainer cap groove and basic at bottom of large cavity	Gage	12013
114.	Wall thickness between slider cavity and angled sides, min	Gage	12014
115.	Width of retainer cap groove, min	Gage	12015
116.	Depth of retainer cap groove, min	Gage	12016
117.	Diameter of propellant cavity	Gage	12017
118.	Depth to bottom of propellant cavity	Gage	12018
119.	Depth to basic at bottom of large cavity	Gage	12019
120.	Location of basic on hinge pin hole countersink	Gage	12020
121.	Symmetry of outside flats above hinge pin holes with large cavity	Gage	12021
122.	Symmetry of basic widths near hinge pin holes with large cavity (2)	Gage	12022
123.	Perpendicularity of slider cavity with large cavity and housing sides	Gage	12023
124.	Positioning of yoke pin holes with relation to large cavity and slider cavity	Gage	12024
125.	Positioning of hinge pin holes with relation to large cavity including perpendicularity with housing sides	Gage	12025
126.	Positioning of slider stop pin hole with relation to large cavity including perpendicularity with slider cavity	Gage	12026

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127.	Diameter of retainer cap groove	Gage	12027
128.	Concentricity of firing pin hole with large cavity	Gage	12028
129.	Burrs or scratches in flash hole, slider cavity or large cavity	Visual	12029
130.	Flash hole missing	Visual	12030
Major	AQL 0.40 Percent (See 4.3.3.2)		
131.	Profile of large cavity	Gage	12031
Minor	AQL 4.00 Percent		
201.	Distance between flats	Gage	12032
202.	Diameter of firing pin hole	Gage	12033
203.	Width of yoke slots	Gage	12034
204.	Depth of yoke pin holes, max	Gage	12035
205.	Depth of hinge pin holes counterbore, max	Gage	12036
206.	Symmetry of yoke slots with housing sides	Gage	12037
207.	Concentricity of flash hole with large cavity	Gage	12038
208.	Diameter of slider cavity counterbore	Gage	12039
209.	Depth of slider cavity counterbore	Gage	12040
210.	Concentricity of slider cavity counterbore with slider cavity	Gage	12041
211.	Concentricity of propellant cavity with large cavity	Gage	12042
212.	Radius or chamfer missing	Visual	12043
213.	Finish improper	Visual	12044
214.	Evidence of poor workmanship (See 3.5)	Visual	12045

*Combined AQL for Major defects 101-131 inclusive, is as follows: AQL 2.50 Percent (See 4.3.2)

4.3.2.13 Diaphragm (See Dwg. C-9211963).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Flange missing	Visual	13001
102.	Part damaged (punctured or torn)	Visual	13002

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Categories	Defects	Method of Inspection	Code No.
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	13003

4.3.2.14 Restrictor (See Dwg. C-9211964).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent (See 4.3.3.4)		
101.	Diameter of flange, min	Gage	14001
102.	Thickness of flange	Gage	14002
103.	Total thickness	Gage	14003
104.	Evidence of poor workmanship (See 3.5)	Visual	14004
Minor:	None defined		

4.3.2.15 Adapter, tube (See Dwg. C-9211968).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.50 Percent		
101.	Diameter of flash hole	Gage	15001
102.	Outside diameter (after knurling), max	Gage	15002
103.	Pitch diameter of thread	Gage	15003
104.	Major diameter of thread, min	Gage	15004
105.	Diameter of counterbore, max	Gage	15005
106.	Diameter of larger cavity (below counterbore)	Gage	15006
107.	Diameter of smaller cavity	Gage	15007
108.	Concentricity of flash hole with pitch diameter of thread	Gage	15008
109.	Concentricity of small cavity with pitch diameter of thread	Gage	15009
110.	Depth to bottom of larger cavity	Gage	15010
111.	Length of large outside diameter including chamfer	Gage	15011
112.	Length of min effective thread	Gage	15012
113.	Concentricity of outside diameter with pitch diameter of thread	Gage	15013

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114.	Concentricity of counterbore with larger cavity	Gage	15014
115.	Concentricity of larger cavity with pitch diameter of thread	Gage	15015
116.	Step missing bottom of counterbore on inside of large inside diameter	Visual	15016
117.	Operation missing	Visual	15017
Minor	AQL 1.50 Percent		
201.	Total length	Gage	15018
202.	Radius or chamfer missing or improper	Visual	15019
203.	Thread damaged	Visual	15020
204.	Evidence of poor workmanship (See 3.5)	Visual	15021

4.3.2.16 Housing assembly (with spirol pin and prior to assembly of yoke and safety wire) (See Dwg. D-9211947).

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Slider in armed position (primer visible)	Visual	16001
Major	AQL 0.65 Percent		
101.	Depth to slider stop, max	Gage	16002
102.	Slider fails to move freely from unarmed to armed position when spirol pin is removed	Manual-Visual	16003
103.	Slider stop missing or above flush	Visual	16004
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	16005

4.3.2.17 Housing assembly (with spirol pin in place, prior to assembly of safety wire (See Dwg. D-9211947).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major	AQL 1.00 Percent		
101.	Height of hinge pin heads from housing, max (4 pins)	Gage	17001
102.	Height of yoke pin heads (2) above housing, max	Gage	17002
103.	Yoke assembly does not move up and down freely	Manual	17003
104.	Outer springs not free to pivot	Manual	17004
105.	Outer springs do not lock properly	Manual	17005
106.	Component missing (including spirol pin)	Visual	17006

Minor: None defined

4.3.2.18 Housing assembly (See Dwg. D-9211947).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Depth to end of slider, max (on red mark side)	Gage	18001
2.	Depth to firing pin, max	Gage	18002
3.	Hinge pin (4) loose	Manual	18003
4.	Hinge pin (4) missing	Visual-Manual	18004
5.	U-bends missing from safety wire	Visual	18005
6.	Safety wire missing	Visual	18006
7.	Safety wire not completely engaged in safety wire hooks	Visual	18007
Major	AQL 1.00 Percent		
101.	Periphery, max	Gage	18008
102.	Safety wire incorrectly positioned	Gage	18009
103.	Red mark missing, on wrong side of housing, or inadequate (too small)	Visual	18010
104.	Component missing (excluding safety wire and hinge pins)	Visual	18011
105.	Component improperly assembled (excluding safety wire and hinge pins)	Visual	18012
106.	Evidence of poor workmanship in large cavity (See 3.5)	Visual	18013

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	107. Spirol pin not removed	Visual	18014
Minor	AQL 1.00 Percent		
	201. Evidence of poor workmanship (See 3.5)	Visual	18015
	202. Coating on yoke damaged (damaged areas not recoated)	Visual	18016

4.3.2.19 Box, inner, sealed (prior to insertion into barrier bag) (See Dwg. F-8865574).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
	101. Box damaged so that contents are exposed	Visual	19001
	102. Sealing strip incomplete, badly wrinkled or missing	Visual	19002
Minor	AQL 1.50 Percent		
	201. Contents loose when box is tilted	Manual	19003
	202. Marking misleading or unidentifiable, including incorrect color code	Visual	19004
	203. Label missing or peeling	Visual	19005

4.3.2.20 Bag, barrier, sealed (prior to insertion into outer box) (See Dwg. F-8865574).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
	101. Waterproofness destroyed by separation of heat seal, or rip, tear or puncture	Visual	20001
Minor	AQL 0.65 Percent		
	201. Marking misleading or unidentifiable	Visual	20002

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4.3.2.21 Box, outer, sealed (See Dwg. C-8865575).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Box damaged so that contents are exposed	Visual	21001
102.	Sealing strip incomplete, badly wrinkled or missing	Visual	21002
Minor	AQL 1.00 Percent		
201.	Contents loose (when box is tilted)	Manual	21003
202.	Marking misleading or unidentifiable, including incorrect color code	Visual	21004

4.3.3 Testing

4.3.3.1 Dimensional control of cast parts.-In place of the normal sampling associated with the classification of defects, and with the approval of the Contracting Officer, a sample of at least three (3) parts (as cast) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production, or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of each 5000 parts. Individual cavity identification must be provided.

If any defective parts are found during qualification of the cavity, the cavity producing the defective parts shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective parts shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

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4.3.3.2 Tool control inspection covering the profile of the large cavity in housing, Code No. 12031.-Tool control may be used in place of the normal sampling associated with the classification of defects and with the approval of the Contracting Officer. At the start of production, after production of each 2000 pieces from each tool, and whenever a change or modification of a form tool is made during production, the required characteristic measurements shall be obtained from that tool. Determination of characteristics through the use of approved plastic or other impressions is permissible. The contractor's tolerance limits for control of characteristics shall be approved by the procuring activity. When a measurement falls outside the contractor's approved tolerance limits, fifty (50) housings from the latter portion of production from that tool, since the last acceptable measurement was taken, shall be measured for the required characteristics. The use of approved impressions is a permissible means of obtaining these characteristics for measurement. If any characteristic measurement taken in this manner falls outside the component drawing tolerance, the housings produced from that tool since the last satisfactory measurement was obtained shall be inspected one hundred percent for that characteristic by the contractor to remove all components in which one or more characteristics are outside drawing tolerance.

In addition, the required characteristic measurements shall be taken from five (5) housings from each eight (8) hour's production, or portion thereof, from each production tool in use. The use of approved impressions is a permissible means of obtaining these characteristics for measurement. If any characteristic measurement falls outside the drawing requirement, the housings produced since the last satisfactory measurement was obtained shall be inspected one hundred percent by the contractor to remove housings which do not comply with the applicable drawing requirement

4.3.3.3 Dimensional control of stamped parts.-In place of the normal sampling associated with the classifications of defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective parts shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

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4.3.3.4 Dimensional control of plastic parts.-In place of the normal sampling associated with the Classifications of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (See Note 1) a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

The inspection of the parts shall be performed only when the curing time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of each 5000 parts or at the end of each shift, providing the number of parts produced during that shift is less than 5000 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

(NOTE 1)- In establishing a curing time, dimensionally check ten (10) parts from each cavity at fifteen (15)-minute intervals until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, SMUPA-ND3. If any new change in material, cycle time, a reworked cavity, or new cavity is used, a new curing time shall be established and approved.

4.3.3.5 Check test for deterioration of primers (See AEI-170) Code No. 22001.-If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into the sliders exceeds two (2) years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous

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tests, the primer lot shall be subjected to, and must satisfactorily pass, the check test specified in AEI-170 immediately before the primer lot is assembled into the slider. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the sliders. (See 6.4)

4.3.3.6 Hardness (Major Defect).-Ten (10) samples of each of the following components shall be selected from each heat treatment batch after completion of all heat treatment operations and subjected to this test. If one or more components fail to comply with the specified hardness requirement, the quantity represented by the failing component shall be rejected. The test shall be performed as specified in 4.4.1. using equipment in accordance with 4.3.4.

COMPONENT	SEE DRAWING	CODE NO.
Slider stop pin	B-9211965	23001
Yoke pin	B-9211954	24001
Hinge pin	C-9211950	25001
Adapter, tube	C-9211968	26001

4.3.3.7 Mechanical properties of housing (prior to machining slider hole) Major Defect (See Dwg. F-9211948).

4.3.3.7.1 Batch or unit heat treatment, Code No. 27001.-Three (3) housings shall be selected from each heat treatment batch after completion of the heat treatment process, and the three (3) specimens therefrom shall be subjected to this test. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.3.7.2 Continuous heat treatment, Code No. 28001.-Three (3) housings shall be selected from each eight (8) hour's production, or portion thereof, after completion of the heat treatment process, and the three (3) specimens prepared therefrom shall be subjected to this test. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.7.3 Rejection.-If any housing specimen fails to comply with the mechanical properties requirements specified on the applicable drawing, the heat treatment quantity represented by the failing specimen shall be rejected.

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4.3.3.8 Permanent deformation of housing (See Dwg. F-9211948) Major Defect, Code No. 29001.-Fifteen (15) housings shall be selected from each lot and subjected to this test. If any housing fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.3, using equipment in accordance with 4.3.4.

4.3.3.9 Push-out test of retainer cap (See Dwg. D-9211971) Major Defect, Code No. 30001.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. Any retainer cap which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.10 Load tests of slider spring (after temperature-conditioning) (See Dwg. C-9211962) Major Defect, Code No. 31001.-Fifty (50) slider springs shall be selected from each lot and subjected to these tests. If any slider spring fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The tests shall be performed as specified in 4.4.5, using equipment in accordance with 4.3.4.

4.3.3.11 Deflection load tests of springs of yoke (See Dwg. D-9211952) Major Defect, Code No. 32001.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. Any yoke which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4.

4.3.3.12 Deflection load test of leaf springs of outer spring (See Dwg. F-9211944) Major Defect, Code No. 33001.-The The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. Any outer spring which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.7. using equipment in accordance with 4.3.4.

4.3.3.13 Push test of side tabs of outer spring (See Dwg. F-9211944) Major Defect, Code No. 34001.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. (This test may be performed on the same sample tested in 4.3.3.12). Any outer spring which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.8, using equipment in accordance with 4.3.4.

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4.3.3.14 Direction of grain of yoke and outer spring (See Dwg. D-9211952 and F-9211944) Major Defect, Code No. 35001.- Three (3) yokes and three (3) outer springs shall be selected from each lot and subjected to this test. The samples selected shall have come from the beginning, middle and end of each roll of raw stock. If any yoke or outer spring fails to comply with the specified grain requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.9, using equipment in accordance with 4.3.4.

4.3.3.15 Leakage test of diaphragm (See Dwg. C-9211963) Major Defect, Code No. 36001.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. Any diaphragm which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.10, using equipment in accordance with 4.3.4.

4.3.3.16 Air flow test of restrictor (See Dwg. C-9211964) Major Defect, Code No. 37001.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. Any restrictor which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.11, using equipment in accordance with 4.3.4.

4.3.3.17 Push test of firing pin of yoke assembly (See Dwg. C-9211951) Major Defect, Code No. 38001.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. Any yoke assembly which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.12 using equipment in accordance with 4.3.4.

4.3.3.18 Push test of primer in slider assembly (See Dwg. B-9211955) Major Defect, Code No. 39001.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter K and an AQL of 0.40 percent. Any slider assembly which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.13, using equipment in accordance with 4.3.4.

4.3.3.19 Security of hinge pins and outer spring in housing assembly (See Dwg. D-9211947) Major Defect, Code No. 40001.- Eighty (80) assemblies shall be selected from each lot and subjected to the falling-weight test specified on the applicable drawing.

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If, during the test, eleven (11) or more holes in the outer springs, or, two (2) or more hinge pins fail to meet the requirements specified on the applicable drawing, the lot shall be rejected. The maximum of ten (10) failures allowed for holes in the outer springs shall contain no more than a total of two (2) failures wherein either a section of metal between the hole and the edge is torn out, or a hole in a corner of either spring has pulled free from the hinge pin. The test shall be performed as specified in 4.4.14, using equipment in accordance with 4.3.4.

4.3.3.20 Pull test of safety wire (See Dwg. D-9211947) Major Defect, Code No. 41001.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter L and an AQL of 0.25 percent. Any housing assembly which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.15, using equipment in accordance with 4.3.4.

4.3.3.21 Delay arming time of housing assembly (See Dwg. D-9211947) Major Defect, Code No. 42001.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Code Letter L and an AQL of 0.65 percent. Any housing assembly which fails to comply with the specified requirement shall be classed defective. The test shall be performed as specified in 4.4.16 using equipment in accordance with 4.3.4.

4.3.3.22 Simulated grenade catapult test (See 3.3) Major Defect, Code No. 43001.-One-hundred and twenty-five (125) housing assemblies shall be selected from each lot and subjected to this test. If three (3) or more housing assemblies fail to comply with the specified requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.17 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9211947 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping,

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storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Hardness.-The specified parts shall be tested for hardness in accordance with ASTM Method Designation: E 18-67 (Part 31). This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.2 Mechanical properties of housing (prior to machining slider hole).-The test specimen shall be prepared as indicated in Figures I and II of this specification and ASTM E 8-69, Part 31, Figure 8. (smallest size specimen). The determinations for yield point, tensile strength and elongation shall be determined in accordance with Method E 8-69. This is a destructive test.

4.4.3 Permanent deformation of housing.-The housing shall be assembled in a fixture as shown on the Equipment List and the load, as specified on Dwg. F-9211948, applied to the housing. Reading for deformation shall be checked after removal of load. This is a destructive test. Parts so tested shall not be returned to the lot. This test shall be performed prior to machining the slider hole.

4.4.4 Push-out test of retainer cap.-The retainer cap shall be assembled in a fixture as shown on the Equipment List and the load specified on Dwg. D-9211971, applied at the indicated rate. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.5 Load tests of slider spring (after temperature-conditioning).-After temperature-conditioning the compressed spring, as specified on Dwg. C-9211962, the spring shall be assembled to the fixture and the loads applied at the two (2) specified heights as indicated on the applicable drawing. These are non-destructive tests. Slider springs so tested may be returned to the lot.

4.4.6 Deflection load test of springs of yoke.-The yoke shall be assembled in a fixture as shown on the Equipment List and the load specified on Dwg. D-9211952 applied to the springs. This is a non-destructive test. Parts so tested may be returned to the lot.

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4.4.7 Deflection load test of leaf springs of outer spring.-The outer spring shall be assembled in a fixture as shown on the Equipment List and the load specified on Dwg. F-9211944 applied to the spring fingers. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.8 Push test of side tabs of outer spring.-The two (2) tabs of the outer spring shall be pushed flush, released and gaged as specified on the applicable drawing. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.9 Direction of grain of yoke and outer spring.-This test shall be conducted by any method satisfactory to the Contracting Officer. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.10 Leakage test of diaphragm.-The diaphragm shall be subjected to the air pressure for the minimum time as specified on Dwg. C-9211963. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.11 Air flow test of restrictor.-The restrictor shall be assembled in a fixture as shown on the Equipment List and subjected to the air flow as specified on Dwg. C-9211964. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.12 Push test of firing pin of yoke assembly.-The yoke assembly shall be assembled in a fixture as shown on the Equipment List and the load specified on Dwg. C-9211951 applied to the firing pin. This is a non-destructive test. Parts so tested may be returned to the lot.

WARNING: The following tests: 4.4.13, 4.4.14, 4.4.15 and 4.4.16 are conducted on assemblies containing a loaded component. Necessary safety precautions shall be observed.

4.4.13 Push test of primer in slider assembly.-The slider assembly shall be assembled in a fixture as shown on the Equipment List and the load specified on Dwg. B-9211955 applied to the primer. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.14 Security of hinge pins and outer springs.-The housing assembly, with slider restrained in the unarmed position, shall be assembled in a fixture as shown on the Equipment List, and the load specified on Dwg. D-9211947 applied to the hinge pins and outer springs. This is a destructive test. Parts so tested shall not be returned to the lot.

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4.4.15 Pull test of safety wire.-The housing assembly with the slider restrained in the unarmed position, shall be assembled in a fixture as shown on the Equipment List and the pull specified on Dwg. D-9211947 applied to the safety wire in the direction as indicated. Examination shall be made for any evidence of the safety wire pulling free of the housing assembly. This is a destructive test. Housing assemblies so tested shall be assembled with new safety wires.

4.4.16 Delay arming time of housing assembly.-The housing assembly, with slider in the unarmed position, shall be assembled in a fixture and tested as specified on the Equipment List. Observation shall be made for movement of the slider assembly into the armed position within the time specified on Dwg. D-9211947. Acceptable housings shall then be disarmed (sliders returned to the unarmed position) in a fixture specified on the Equipment List and returned to the lot. This disarming shall be accomplished using a max. pressure of eighty (80) psi. Assemblies that do not pass the arming time test shall not be disarmed nor returned to the lot.

4.4.17 Simulated grenade catapult test.-Grenades shall be modified (See 6.5), loaded and assembled using equipment specified on EL 9211947. The grenade shall be assembled with a Government propelling charge and a ball to simulate the weight and size as specified in 3.3. Assembly shall be in accordance with Dwg. C-9211946. Instrumentation shall be provided to check impact velocity on every fifth sample fired. Velocity results shall be recorded.

4.4.17.1 Results to be discarded.-Recorded data required by 4.4.17.2 shall be discarded if any one of the following occurs:

a. Impact velocity less than 100 feet per second.
b. Equipment malfunction
c. Poor orientation on impact
d. External interference to ball while in flight

(Additional grenades shall be provided to complete the sample quantity specified).

4.4.17.2 Results to be recorded.-Record, for informational use, the following:

a. Impact velocity
b. Housing malfunctions
c. Crushed or cracked housings

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (Level C).

5.1.1 Tube adapters and retainer caps.-Tube adapters and retainer caps shall be preserved and packaged in accordance with MIL-STD-1169.

5.1.2 Housing assemblies.-The housing assemblies shall be preserved and packaged in accordance with Dwg. F-8865574.

5.2 Packing (Level C).

5.2.1 Tube adapters and retainer caps.-Tube adapters and retainer caps, preserved and packaged as specified in 5.1.1, shall be packed in accordance with MIL-STD-1169.

5.2.2 Housing assemblies.-The housing assemblies, preserved and packaged as specified in 5.1.2, shall be packed in accordance with Dwg. C-8865575.

5.3 Marking.

5.3.1 Tube adapters and retainer caps.-Marking of the boxes shall comply with MIL-STD-1169.

5.3.2 Housing assemblies.-Marking of the boxes shall comply with Dwgs. F-8865574, C-8865575 and Code of Federal Regulations, Title 49, Parts 171-179.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

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6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contrac-

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tor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procurement of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Cost of check test.-The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense

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without cost to the contractor who loaded the primer assemblies or to the contractor assembling the primers into the slider, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer assembly lot was under the jurisdiction of either contractor.

6.5 Preparation of housing assembly for functioning test.- The housing assembly (Dwg. 9211947) selected for this test shall be modified as follows: A standard one-sixteenth (.0625) inch hole shall be drilled through the restrictor, centering the drill 1/32 (0.0312) of an inch from the outer edge of the restrictor. The hole shall be drilled to a depth of 1/8 (0.125) inch. CAUTION: Care shall be taken to prevent chips from entering the hole by the use of compressed air, or other suitable means, during the drilling operation.

The item shall then be loaded and tested as specified in 4.4.17.

6.6 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

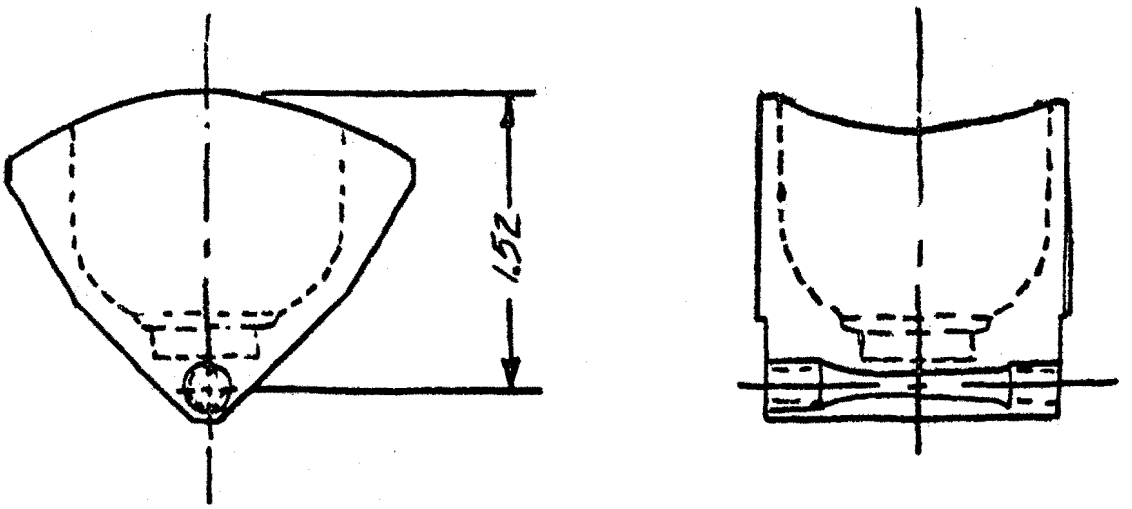
6.7 Supersession data.-This specification includes the requirements of, and supersedes, Purchase Specification AEI-390, dated 21 February 1968.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1330-A 001

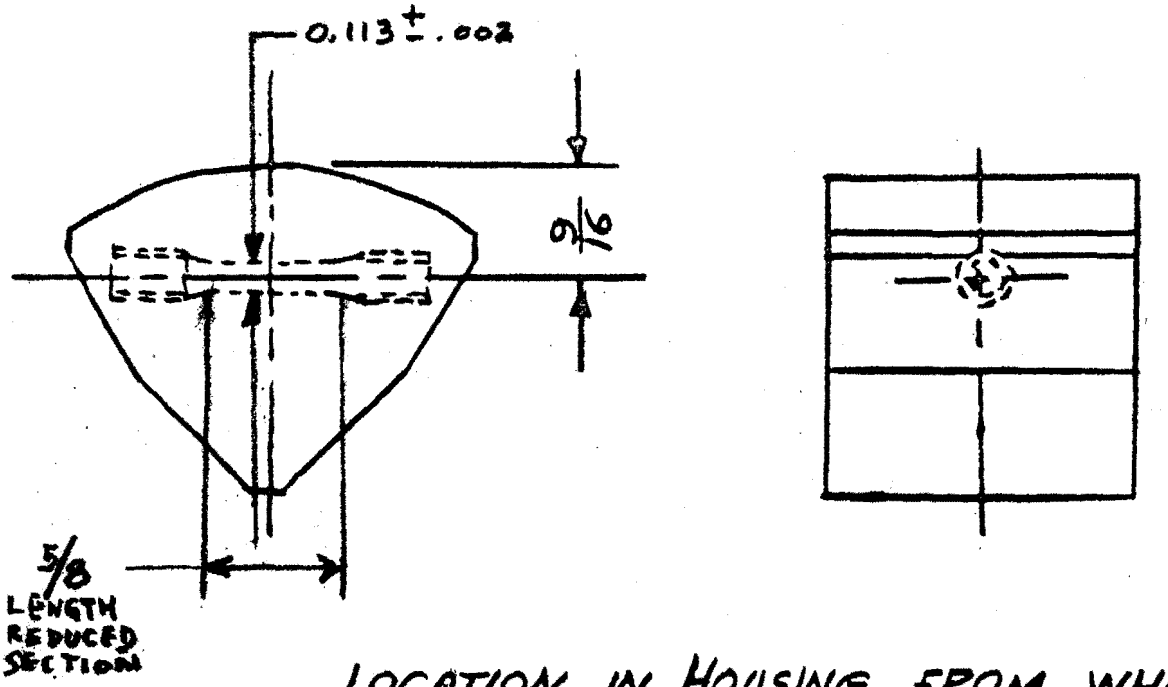
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LOCATION IN HOUSING FROM WHICH TENSILE SPECIMEN IS TAKEN (FOR GROSS FORGING)

FIGURE 1

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LOCATION IN HOUSING FROM WHICH TENSILE SPECIMEN IS TAKEN (FOR EXTRUDED BAR STOCK)

FIGURE 2

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